

Project Report

For the Course

EE-313 Electronic Circuit Design



For

B.E. Electrical Engineering

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Audio Amplifier Project Report

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1. Objectives

This project aims to design, simulate, and validate a complete discrete-component audio amplifier using MOSFETs and BJTs without the use of integrated circuits. The key objectives are:

- Design a Class AB push-pull output stage to deliver 3–5 W into an 8 Ω loudspeaker with low crossover distortion.
- Amplify millivolt-level signals from an electret condenser microphone using a multi-stage pre-amplifier with high input impedance.
- Simulate the complete circuit in LTspice to verify gain, frequency response, and power delivery at each stage.
- Ensure proper impedance matching between all stages to maximise signal transfer and prevent loading effects.
- Achieve a mid-band voltage gain of approximately 500–700 V/V and a bandwidth of 100 Hz – 20 kHz.

2. Equipment Used

Component	Type / Model	Purpose
MOSFET (M1)	2N7000	Pre-amplifier (first stage)
NPN BJT (Q5)	BC547 / 2N3904	Pre-amplifier / gain stage
PNP BJT (Q3)	General PNP	Driver stage
Power NPN (Q1)	TIP41	Output stage (upper half)
Power PNP (Q4)	TIP42	Output stage (lower half)
Diodes (D1–D4)	1N4007	Crossover distortion reduction
Resistors	Various	Biasing, gain, and feedback
Capacitors	Various	Coupling, bypass, compensation
Loudspeaker	8 Ω	Output load
Microphone	Electret condenser	Input signal source
Power Supply	24 V DC	Circuit supply
Heat Sinks	—	Cooling for power transistors

3. Main stage: Class AB Power Amplifier (Output Stage)

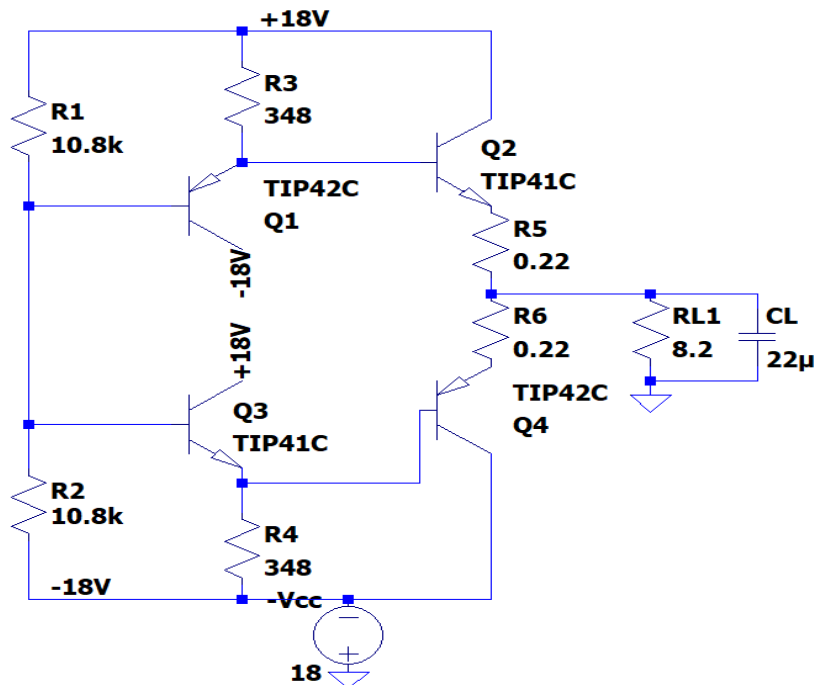
3.1 Introduction

The Class AB power amplifier is the final output stage of the audio amplifier. Its primary role is to deliver 3–5 W of audio power to an 8 Ω loudspeaker while maintaining low distortion and good efficiency. A complementary push-pull transistor pair (TIP122 NPN and TIP127 PNP) is used to ensure full-cycle signal coverage, while diode biasing between the transistor bases eliminates crossover distortion.

3.2 Design Procedure

- Selected Class AB push-pull configuration with complementary transistors (TIP122 NPN and TIP127 PNP).
- Applied diode biasing (D1–D4, 1N4007) between transistor bases to reduce crossover distortion.
- Added small emitter resistors for current sharing and thermal stability.
- Calculated supply voltage using: $V_{cc} \approx \sqrt{(8 \times P \times R_L)} + V_{CE(sat)}$; result: $V_{cc} \approx 24$ V.
- Designed for 8 Ω speaker load with target output power of 3–5 W.
- Simulated in LTspice to verify ± 5 V output swing and ~ 1.5 A peak current drive.
- Input impedance of output stage (~ 820 Ω) matched to the preceding driver stage.

3.3 Calculations



Assuming $\beta = 30$ for TIP 41C, TIP 42C

Assuming $V_{EB} = 0.6V$

$$i_L = \frac{5}{8.2} = 0.6098 A \approx \mathbf{0.61 A}$$

So As only one of Q_2 and Q_4 are on at a time so they must supply 0.61 A during their on cycle.

$$ie_{Q2} = 0.61A$$

$$ib_{Q2} = \frac{0.61}{30} A$$

$$= \mathbf{19.7mA}$$

By rule of thumb for Q_1 to supply 19.7mA,

$$ic_{Q1} = 2.5 \times ib_{Q2}$$

$$= 49.25mA \approx 50mA$$

$$ib_{Q1} = \frac{50m}{30} A$$

$$= 1.667mA$$

$$R_3 = \frac{18 - V_{EB}}{50m} = \mathbf{348\Omega}$$

Similarly for Q_2 and Q_4

$$ie_{Q4} = 0.61A$$

$$ib_{Q2} = \mathbf{19.7mA}$$

$$ic_{Q2} = 2.5 \times ib_{Q4} = 49.25mA \approx 50mA$$

$$R_4 = \frac{-0.6 - (-18)}{50mA} = \mathbf{348\Omega}$$

$$ib_{Q2} = \frac{50m}{30} A$$

$$= 1.667mA$$

$$R_1 = R_2 = \frac{18}{1.667m} = 10.797k\Omega$$

R_5 and R_6 are chosen to reduce effects of V_{BE} and V_{EB} mismatches and thermal runaway and taken as 0.22Ω each based on standards.

$$R_5 = R_6 = 0.22\Omega$$

$$R_{in} = 10.8k \parallel R_B$$

$$R_B = r_\pi + (\beta + 1)R_E$$

$$R_B = (\beta + 1)r_e + (\beta + 1)R_E$$

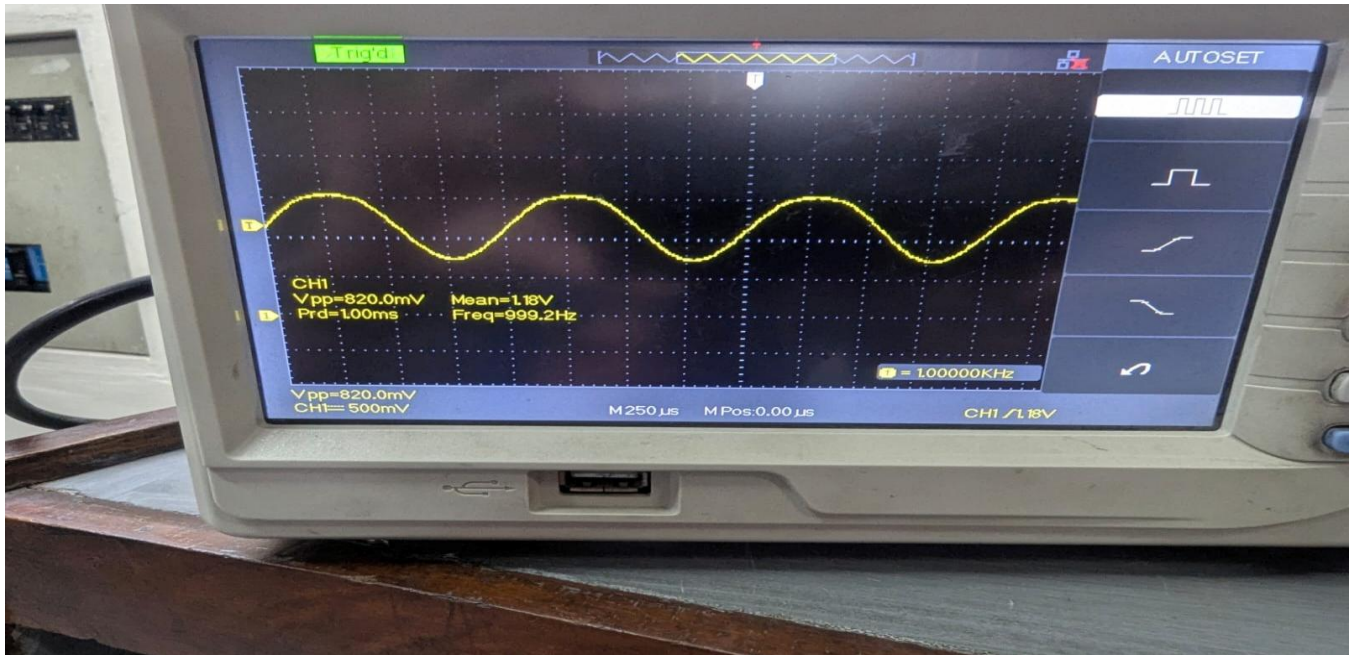
$$re \approx \frac{25 mV}{I_E} = \frac{25 mV}{48.39mA} = 0.517$$

$$R_B = (31 \times 0.517) + (31 \times 348) = 10.805k\Omega$$

$$R_{in} = 10.8k \parallel (10.805k) = \mathbf{5.4k\Omega}$$

$$R_{in} = R_{L \text{ of prev stage}} = \mathbf{5.4k\Omega}$$

Stage 1 Waveforms



4. Stage 2: PNP Driver Stage

Introduction

The PNP driver stage (Q3) amplifies the signal from the pre-amplifier and provides sufficient current gain and voltage swing to drive the Class AB output stage. Operating in common-emitter configuration, it ensures the preceding stage is not loaded while delivering a clean, amplified signal to the power transistors.

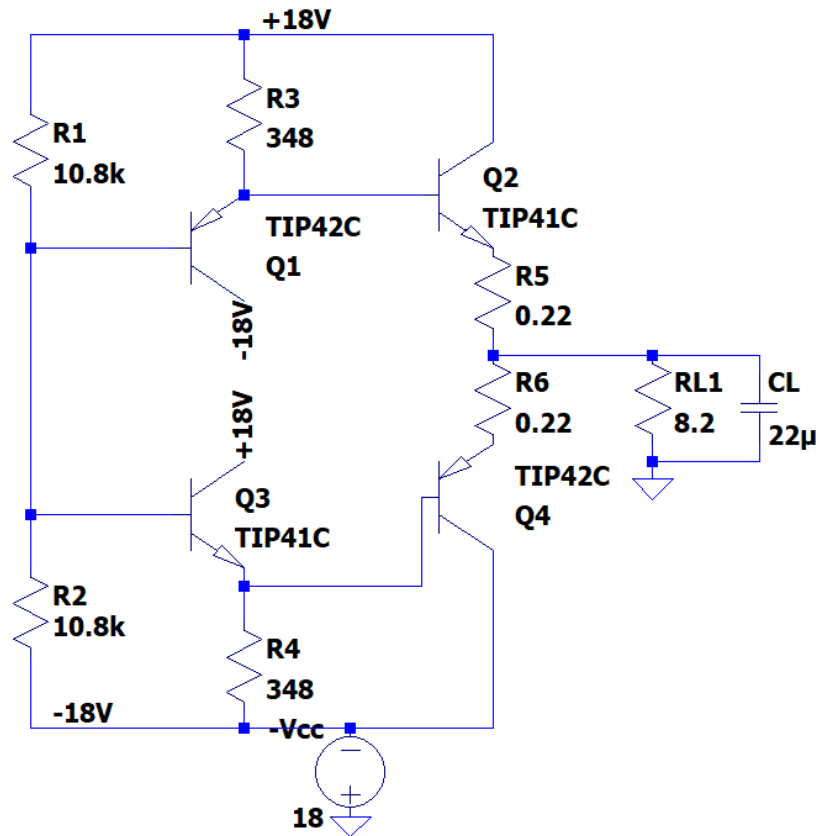
Design Procedure

- Used PNP transistor Q3 in common-emitter configuration.
- Base bias resistors: $R_{10} = 2.88 \text{ k}\Omega$, $R_{11} = 1.93 \text{ k}\Omega$ (voltage divider).
- Emitter resistor $R_9 = 343 \text{ }\Omega$ for DC stability; bypassed with $C_4 = 22 \text{ }\mu\text{F}$ for AC gain.
- Collector load resistor $R_7 = 1.714 \text{ k}\Omega$.
- Input coupling capacitor $C_2 = 10 \text{ }\mu\text{F}$.
- Simulated in LTspice to verify gain and phase response.
- Input impedance $\approx 1.56 \text{ k}\Omega$, matched to MOSFET + BJT pre-amplifier output.

Design Values Summary

Parameter	Symbol	Value
Supply Voltage	V_{cc}	24 V
Collector Current	I_C	7 mA
Current Gain (assumed)	β	100
Emitter Voltage	V_E	2.4 V
Base Voltage	V_B	1.7 V
Emitter Resistor	R_E	343 Ω
Collector Resistor	R_C	1.714 k Ω
Input Impedance	R_{in}	≈ 1.56 k Ω

Calculations



Assuming $\beta = 30$ for TIP 41C, TIP 42C

Assuming $V_{EB} = 0.6V$

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$$ie_{Q2} = 0.61A$$

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$$= \mathbf{19.7mA}$$

By rule of thumb for Q_1 to supply 19.7mA,

$$ic_{Q1} = 2.5 \times ib_{Q2}$$

$$= 49.25mA \approx 50mA$$

$$ib_{Q1} = \frac{50m}{30} A$$

$$= 1.667mA$$

$$R_3 = \frac{18 - V_{EB}}{50m} = \mathbf{348\Omega}$$

Similarly for Q_2 and Q_4

$$ie_{Q4} = 0.61A$$

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$$R_4 = \frac{-0.6 - (-18)}{50mA} = \mathbf{348\Omega}$$

$$ib_{Q2} = \frac{50m}{30} A$$

$$= 1.667mA$$

$$R_1 = R_2 = \frac{18}{1.667m} = 10.797k\Omega$$

R_5 and R_6 are chosen to reduce effects of V_{BE} and V_{EB} mismatches and thermal runaway and taken as 0.22Ω each based on standards.

$$R_5 = R_6 = 0.22\Omega$$

$$R_{in} = 10.8k \parallel R_B$$

$$R_B = r_\pi + (\beta + 1)R_E$$

$$R_B = (\beta + 1)r_e + (\beta + 1)R_E$$

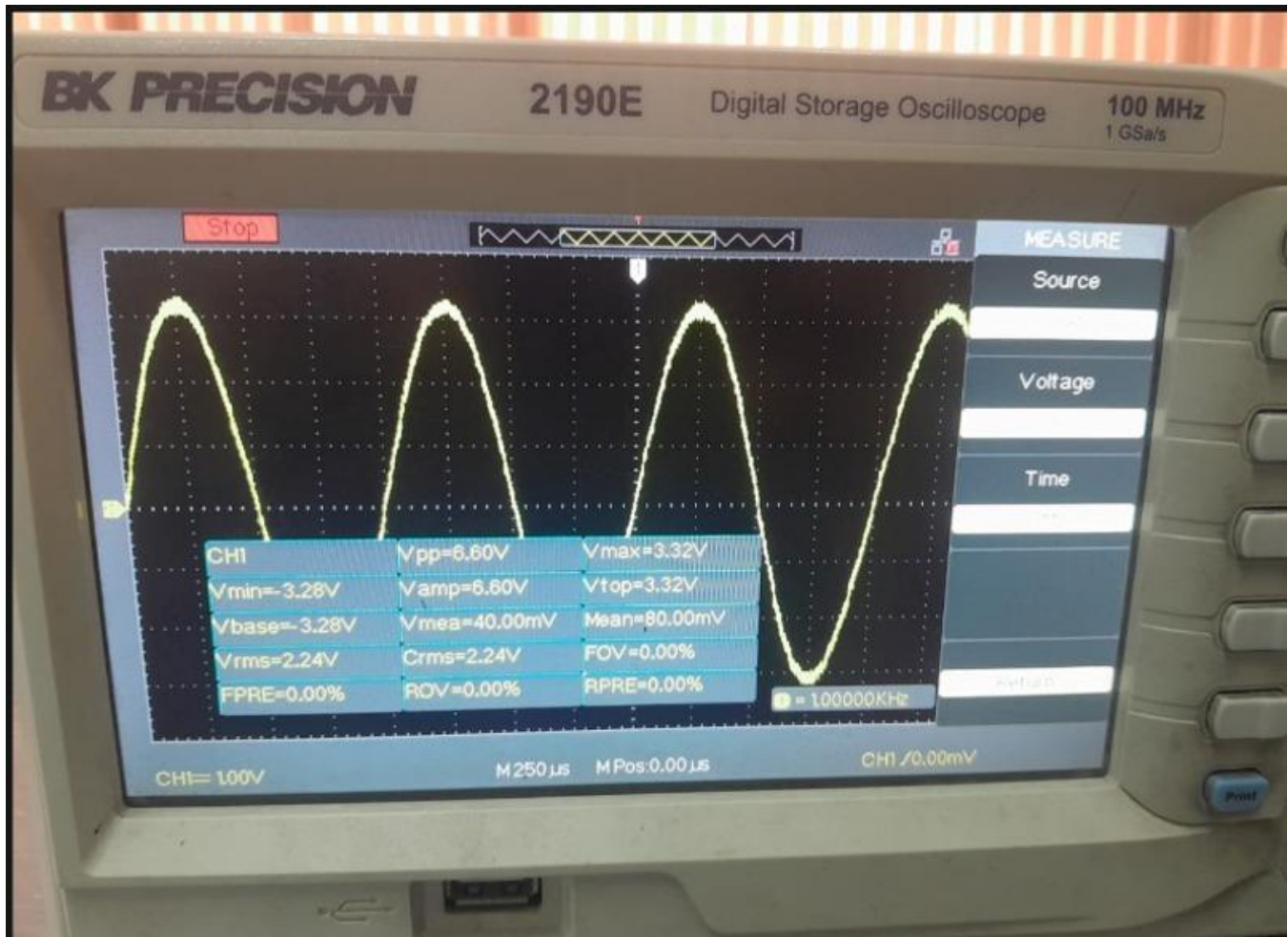
$$re \approx \frac{25 mV}{I_E} = \frac{25 mV}{48.39mA} = 0.517$$

$$R_B = (31 \times 0.517) + (31 \times 348) = 10.805k\Omega$$

$$R_{in} = 10.8k \parallel (10.805k) = \mathbf{5.4k\Omega}$$

$$R_{in} = R_{L \text{ of prev stage}} = \mathbf{5.4k\Omega}$$

Stage 2 Waveforms



5. Stage 3: Combined PNP Driver & Class AB Power Amplifier Introduction

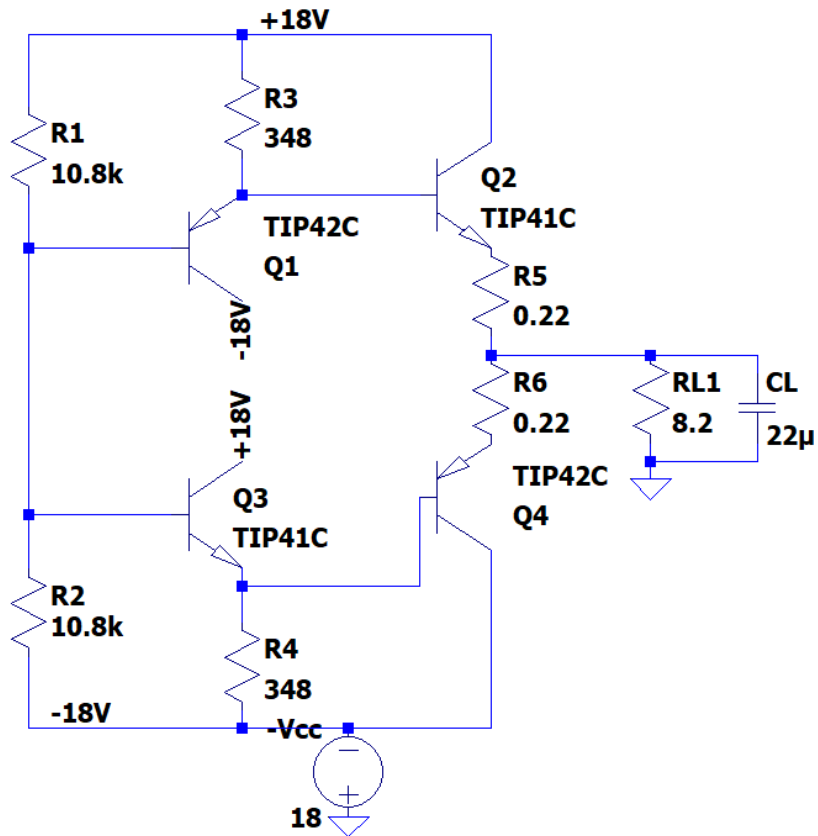
This combined stage integrates the PNP driver (Q3) and the Class AB output stage (Q1, Q2, Q4) into a single high-power amplifier block. The PNP driver boosts current and voltage from the pre-amplifier, while the Class AB stage delivers 3–5 W into the 8 Ω speaker load with low distortion and high efficiency.

Integration Procedure

- PNP driver output (collector of Q3) feeds directly into the bias network of the Class AB stage.
- Q2 (2N3904) used for bias current control of the output transistors.
- Diodes D1–D4 set quiescent bias point of Q1 and Q4.

- Emitter resistors R2, R3 $\approx 0.5 \Omega$ each for thermal current sharing.
- Output capacitor C1 = 220 μ F blocks DC from the speaker.
- R6 = 30 Ω , R7 = 600 Ω used for Q2 biasing.
- C3 added in feedback path for stability.
- Simulated combined circuit in LTspice, verified ± 5 V swing, 3–5 W output, low THD.

Calculations



Assuming $\beta = 30$ for TIP 41C, TIP 42C

Assuming $V_{EB} = 0.6V$

$$i_L = \frac{5}{8.2} = 0.6098 A \approx \mathbf{0.61 A}$$

So As only one of Q₂ and Q₄ are on at a time so they must supply 0.61 A during their on cycle.

$$i_{eQ2} = 0.61A$$

$$i_{bQ2} = \frac{0.61}{30} A$$

$$= \mathbf{19.7mA}$$

By rule of thumb for Q₁ to supply 19.7mA,

$$\begin{aligned}
i_{CQ1} &= 2.5 \times i_{BQ2} \\
&= 49.25mA \approx 50mA \\
i_{BQ1} &= \frac{50m}{30} A \\
&= 1.667mA \\
R_3 &= \frac{18 - V_{EB}}{50m} = \mathbf{348\Omega}
\end{aligned}$$

Similarly for Q₂ and Q₄

$$\begin{aligned}
i_{EQ4} &= 0.61A \\
i_{BQ2} &= \mathbf{19.7mA} \\
i_{CQ2} &= 2.5 \times i_{BQ4} = 49.25mA \approx 50mA \\
R_4 &= \frac{-0.6 - (-18)}{50mA} = \mathbf{348\Omega} \\
i_{BQ2} &= \frac{50m}{30} A \\
&= 1.667mA \\
R_1 = R_2 &= \frac{18}{1.667m} = 10.797k\Omega
\end{aligned}$$

R₅ and R₆ are chosen to reduce effects of V_{BE} and V_{EB} mismatches and thermal runaway and taken as 0.22Ω each based on standards.

$$\begin{aligned}
R_5 &= R_6 = 0.22\Omega \\
R_{in} &= 10.8k \parallel R_B \\
R_B &= r_{\pi} + (\beta + 1)R_E \\
R_B &= (\beta + 1)r_e + (\beta + 1)R_E \\
r_e &\approx \frac{25 mV}{I_E} = \frac{25 mV}{48.39mA} = 0.517 \\
R_B &= (31 \times 0.517) + (31 \times 348) = 10.805k\Omega \\
R_{in} &= 10.8k \parallel (10.805k) = \mathbf{5.4k\Omega} \\
R_{in} &= R_{L \text{ of prev stage}} = \mathbf{5.4k\Omega}
\end{aligned}$$

6. Stage 4: BJT Pre-Amplifier Stage (M1)

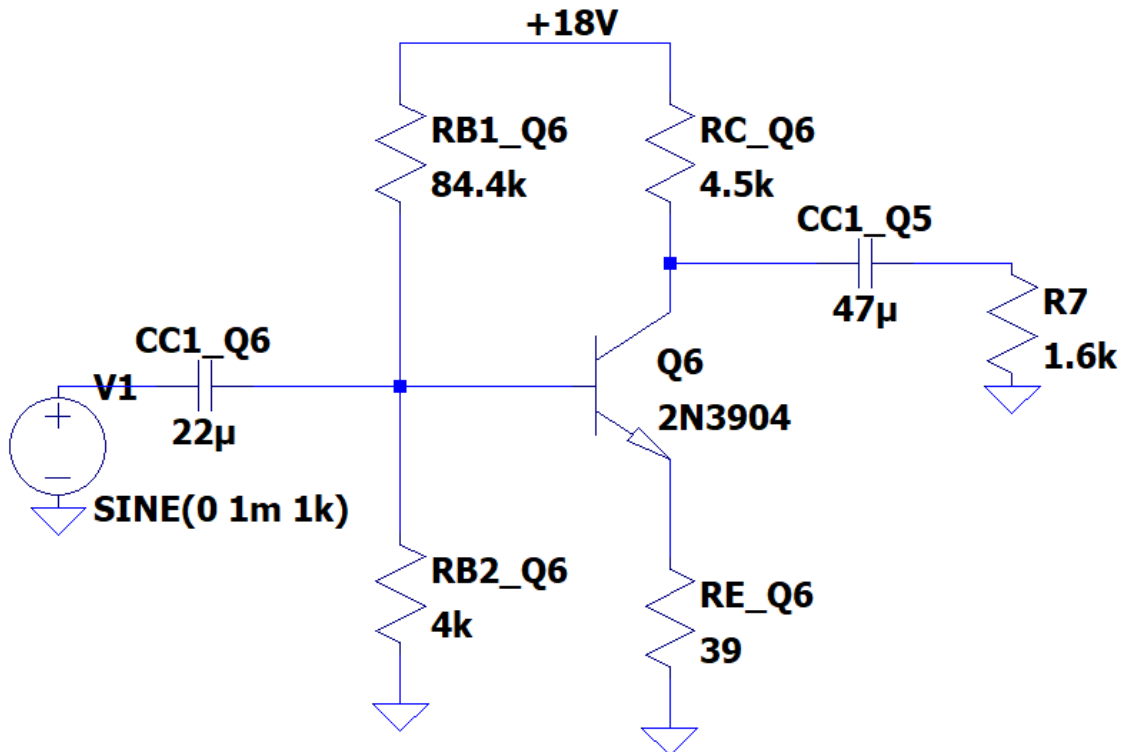
Introduction

The MOSFET pre-amplifier (M1) is the first stage of the audio chain. It receives the very weak millivolt-level signal from the electret condenser microphone and provides an initial voltage gain. The MOSFET (2N7000) is chosen for its very high input impedance, which prevents loading of the microphone source while delivering a usable signal to subsequent stages.

Design Procedure

- Used M1 (2N7000) in common-source configuration.
- Set drain current $I_D = 2$ mA for linear operation in the active region.
- Drain resistor $R_D = R17 = 3500\ \Omega$ to set $V_D = 17$ V.
- Source resistor $R_S = R20 = 300\ \Omega$ for gate bias stability.
- Gate bias network: $R18 = 80.4\ \text{k}\Omega$, $R19 = 39.6\ \text{k}\Omega$; gate voltage $V_G = 7.92$ V.
- Coupling capacitor $C8 = 10\ \mu\text{F}$ at input, $C7 = 100\ \mu\text{F}$ at output.
- Simulated in LTspice to verify gain and input/output impedance.

Calculations



Given

$$V_{CC} = 18 \text{ V}$$

$$I_C = 2 \text{ mA}$$

$$\beta = 80$$

$$R_L = 1.6 \text{ k}\Omega$$

Target voltage gain:

$$A_v \approx 23\text{--}25 \text{ V/V}$$

Collector Voltage Selection

For maximum symmetrical output swing:

$$V_C = \frac{V_{CC}}{2} = 9 \text{ V}$$

Collector Resistor R_C

$$R_C = \frac{V_{CC} - V_C}{I_C}$$
$$R_C = \frac{18 - 9}{2 \times 10^{-3}}$$
$$R_C = 4.5 \text{ k}\Omega$$

Emitter Current I_E

$$\alpha = \frac{\beta}{\beta + 1} = \frac{80}{81}$$
$$I_E = \alpha I_C = \frac{80}{81} \times 2 \text{ mA}$$
$$I_E \approx 1.98 \text{ mA}$$

Small-Signal Emitter Resistance r_e

$$r_e = \frac{25 \text{ mV}}{I_E}$$
$$r_e = \frac{25 \times 10^{-3}}{1.98 \times 10^{-3}}$$
$$r_e \approx 12.6 \Omega$$

Effective Collector Load

$$R_C \parallel R_L = 4.5 \text{ k} \parallel 1.6 \text{ k}$$
$$R_C \parallel R_L = \frac{(4.5)(1.6)}{4.5 + 1.6}$$
$$R_C \parallel R_L \approx 1.18 \text{ k}\Omega$$

Emitter Resistance from Gain Requirement

Voltage gain of a CE stage with emitter degeneration:

$$A_v \approx \frac{R_C \parallel R_L}{r_e + R_E}$$

Rearranging to find R_E :

$$r_e + R_E = \frac{R_C \parallel R_L}{A_v}$$

Taking $A_v \approx 23$:

$$r_e + R_E = \frac{1180}{23} \approx 51.3 \Omega$$

$$R_E = 51.3 - 12.6$$

$$R_E \approx 38.7 \Omega \approx 39 \Omega$$

Emitter Voltage V_E

$$V_E = I_E \times R_E$$

$$V_E = 1.98 \times 10^{-3} \times 39$$

$$V_E \approx 0.077 \text{ V}$$

Base Voltage V_B

$$V_B = V_E + V_{BE}$$

$$V_B = 0.077 + 0.7$$

$$V_B \approx 0.78 \text{ V}$$

Base Current

$$I_B = \frac{I_C}{\beta}$$

$$I_B = \frac{2 \times 10^{-3}}{80}$$

$$I_B = 25 \mu\text{A}$$

Bias Divider Current Selection

Choose a stiff divider:

$$I_{\text{div}} \approx 8-10 I_B$$

Taking:

$$I_{\text{div}} \approx 200 \mu\text{A}$$

Bias Resistor Values

Lower resistor:

$$R_{B2} = \frac{V_B}{I_{\text{div}}}$$

$$R_{B2} = \frac{0.78}{200 \times 10^{-6}}$$

$$R_{B2} \approx 3.9 \text{ k}\Omega \approx 4 \text{ k}\Omega$$

Upper resistor:

$$R_{B1} = \frac{V_{CC} - V_B}{I_{\text{div}}}$$

$$R_{B1} = \frac{18 - 0.78}{200 \times 10^{-6}}$$

$$R_{B1} \approx 84.4 \text{ k}\Omega$$

Input Impedance of CE1 Stage

Small-signal base resistance of the transistor is given by:

$$r_{\pi} = \beta r_e$$

$$r_{\pi} = 80 \times 12.6$$

$$r_{\pi} \approx 1008 \Omega$$

The effective resistance looking into the base due to emitter degeneration is:

$$R_B = r_{\pi} + (\beta + 1)R_E$$

$$R_B = 1008 + 81 \times 39$$

$$R_B = 1008 + 3159$$

$$R_B \approx 4167 \Omega \approx 4.17 \text{ k}\Omega$$

The total input impedance of the CE1 stage is the parallel combination of the biasing resistors and the transistor input resistance:

$$R_{in} = R_{B1} \parallel R_{B2} \parallel R_B$$

$$R_{in} = 84.4k \parallel 4k \parallel 4.17k$$

First combine R_{B2} and R_B :

$$4k \parallel 4.17k = \frac{(4)(4.17)}{4 + 4.17}$$

$$= \frac{16.68}{8.17}$$

$$\approx 2.04 \text{ k}\Omega$$

Now include R_{B1} :

$$R_{in} = 84.4k \parallel 2.04k$$

$$R_{in} \approx 1.99 \text{ k}\Omega$$

7. Stage 5: BJT Pre-Amplifier Stage (Q5)

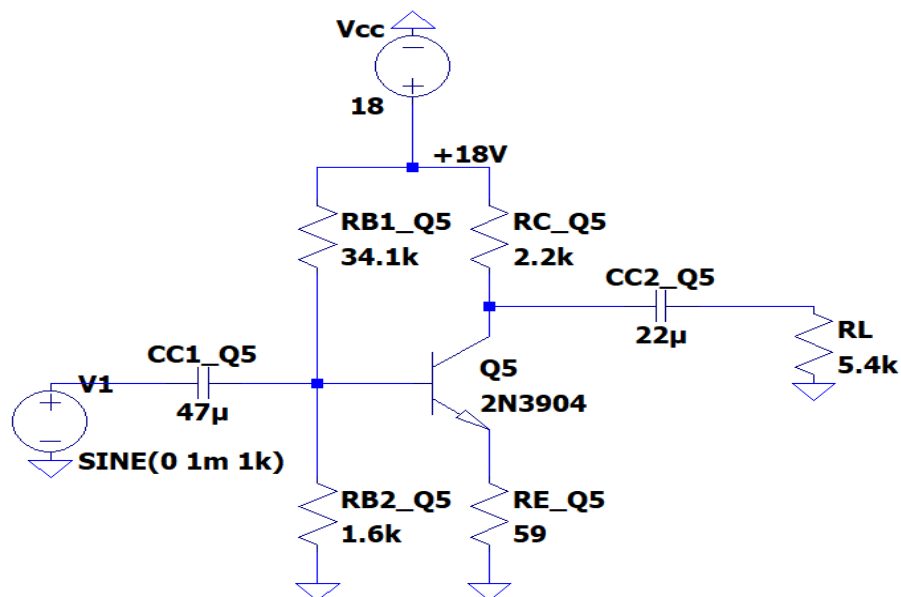
Introduction

The BJT pre-amplifier stage (Q5) provides additional voltage gain after the MOSFET stage. A common-emitter NPN configuration (BC547) is used to further amplify the signal while maintaining good linearity. An unbypassed emitter resistor introduces local negative feedback, which linearises the gain and extends the bandwidth.

Design Procedure

- Used NPN BJT Q5 (BC547) in common-emitter configuration.
- Set collector current $I_C = 3.3 \text{ mA}$, supply $V_{CC} = 24 \text{ V}$, $\beta = 100$.
- Collector resistor $R_C = R_{12} = 4 \text{ k}\Omega$, setting $V_C = 12 \text{ V}$ for maximum swing.
- Emitter resistor $R_E = 727 \Omega$ total; $R_{15} = 18 \Omega$ left unbypassed for AC negative feedback.
- Base bias network: $R_{13} = 41.8 \text{ k}\Omega$, $R_{16} = 700 \Omega$; $V_B = 3.1 \text{ V}$.
- Coupling capacitor $C_6 = 470 \mu\text{F}$ for AC signal transfer at input.
- Voltage gain $A_v = -R_C \parallel R_{in,next} / (r_e' + R_E) \approx -43.8 \text{ V/V}$.
- Input impedance $R_{in} \approx 543 \Omega$.
- Simulated in LTspice to verify gain, frequency response, and distortion levels.

Calculations



Given

$$V_{CC} = 18 \text{ V}$$

$$I_C = 4 \text{ mA (assumed)}$$

$$\beta = 80 \text{ (assumed)}$$

$$R_L = 5.4 \text{ k}\Omega$$

Collector Voltage Assumption

Assume the collector voltage at mid-supply for maximum symmetrical swing:

$$V_C = \frac{V_{CC}}{2} \approx 9 \text{ V}$$

Collector Resistor R_C

$$R_C = \frac{V_{CC} - V_C}{I_C}$$

$$R_C = \frac{18 - 9}{4 \times 10^{-3}}$$

$$R_C = 2.25 \text{ k}\Omega \approx 2.2 \text{ k}\Omega$$

(Selected standard value)

Emitter Current I_E

$$\alpha = \frac{\beta}{\beta + 1} = \frac{80}{81}$$

$$I_E = \alpha I_C = \frac{80}{81} \times 4 \text{ mA} = 3.95 \text{ mA}$$

Small-Signal Emitter Resistance r_e

$$r_e = \frac{25 \text{ mV}}{I_E}$$

$$r_e = \frac{25 \times 10^{-3}}{3.95 \times 10^{-3}}$$

$$r_e \approx 6.33 \Omega$$

Effective Collector Load

$$R_C \parallel R_L = 2.2 \text{ k} \parallel 5.4 \text{ k}$$

$$R_C \parallel R_L = \frac{(2.2)(5.4)}{2.2 + 5.4}$$

$$R_C \parallel R_L \approx 1.56 \text{ k}\Omega$$

Emitter Resistance from Gain Requirement

Voltage gain of a CE stage with emitter degeneration is:

$$A_v \approx \frac{R_C \parallel R_L}{r_e + R_E}$$

Rearranging to find R_E :

$$r_e + R_E = \frac{R_C \parallel R_L}{A_v}$$

$$r_e + R_E = \frac{1560}{24} = 65 \, \Omega$$

$$R_E = 65 - 6.33$$

$$R_E \approx 58.7 \, \Omega \approx 59 \, \Omega$$

Emitter Voltage V_E

$$V_E = I_E \times R_E$$

$$V_E = 3.95 \times 10^{-3} \times 59$$

$$V_E \approx 0.233 \, \text{V}$$

Base Voltage V_B

$$V_B = V_E + V_{BE}$$

$$V_B = 0.233 + 0.7$$

$$V_B \approx 0.93 \, \text{V}$$

Base Current

$$I_B = \frac{I_C}{\beta}$$

$$I_B = \frac{4 \times 10^{-3}}{80}$$

$$I_B = 50 \, \mu\text{A}$$

Bias Divider Current

Choose:

$$I_{\text{div}} = 10I_B = 500 \, \mu\text{A}$$

Bias Resistor Values

Lower resistor:

$$R_{B2} = \frac{V_B}{I_{\text{div}}}$$

$$R_{B2} = \frac{0.93}{500 \times 10^{-6}}$$

$$R_{B2} \approx 1.86 \, \text{k}\Omega$$

Upper resistor:

$$R_{B1} = \frac{V_{CC} - V_B}{I_{\text{div}}}$$

$$R_{B1} = \frac{18 - 0.93}{500 \times 10^{-6}}$$

$$R_{B1} \approx 34.1 \text{ k}\Omega$$

Input Resistance of CE Stage

$$r_{\pi} = \beta r_e = 80 \times 6.33 = 506 \Omega$$

$$R_B = r_{\pi} + (\beta + 1)R_E$$

$$R_B = 506 + 81 \times 59$$

$$R_B \approx 5.29 \text{ k}\Omega$$

Total input resistance:

$$R_{in} = R_{B1} \parallel R_{B2} \parallel R_B$$

$$R_{in} = 34.1k \parallel 1.86k \parallel 5.29k$$

$$R_{in} \approx 1.6 \text{ k}\Omega$$

8. Stage 6: Combined Pre-Amplifier

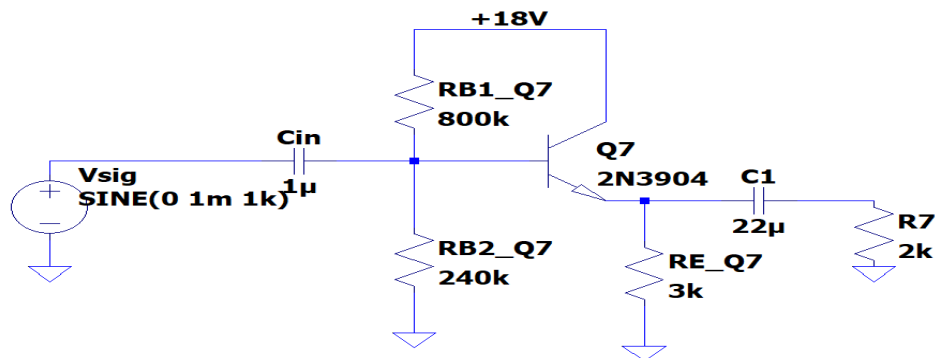
Introduction

The combined pre-amplifier integrates the MOSFET stage (M1) and the BJT stage (Q5) into a single high-gain, high-input-impedance block. The MOSFET provides the critical high-impedance interface to the microphone, while the BJT adds the bulk of the voltage gain. A resistive-capacitive feedback path from the BJT collector to the MOSFET gate is used to set a closed-loop gain ≈ 100 and improve stability and bandwidth.

Cascaded Configuration

- Output of MOSFET stage (M1 drain) is AC-coupled via capacitor C3 to the base of Q5.
- Closed-loop gain ≈ 100 V/V set by feedback network from Q5 collector to M1 gate/source.
- Frequency response verified over 100 Hz – 20 kHz in LTspice simulation.
- Input to the cascade: 10 mV sine at 1 kHz (representing microphone signal).

Calculations



Given

$$V_{CC} = 18 \text{ V}$$

$$I_C = 1 \text{ mA}$$

$$\beta = 80$$

$$R_L = 2 \text{ k}\Omega$$

Emitter Current I_E

For a CC stage:

$$\alpha = \frac{\beta}{\beta + 1} = \frac{80}{81}$$

$$I_E = \alpha I_C = \frac{80}{81} \times 1 \text{ mA}$$

$$I_E \approx 0.988 \text{ mA} \approx 1 \text{ mA}$$

Bias Divider Voltage

Base voltage is set by the divider:

$$V_B = V_{CC} \frac{R_{B2}}{R_{B1} + R_{B2}}$$

$$V_B = 18 \times \frac{220k}{750k + 220k}$$

$$V_B = 18 \times \frac{220}{970}$$

$$V_B \approx 4.08 \text{ V}$$

Emitter Voltage V_E

$$V_E = V_B - V_{BE}$$

$$V_E = 4.08 - 0.7$$

$$V_E \approx 3.38 \text{ V}$$

Emitter Resistor R_E

$$R_E = \frac{V_E}{I_E}$$

$$R_E = \frac{3.38}{0.988 \times 10^{-3}}$$

$$R_E \approx 3.42 \text{ k}\Omega \approx 3 \text{ k}\Omega$$

(Standard value selected)

Base Current

$$I_B = \frac{I_C}{\beta}$$

$$I_B = \frac{1 \times 10^{-3}}{80}$$

$$I_B = 12.5 \mu\text{A}$$

Bias Divider Current Check

$$I_{\text{div}} = \frac{V_{CC}}{R_{B1} + R_{B2}} = \frac{18}{970k}$$

$$I_{\text{div}} \approx 18.6 \mu\text{A}$$

$$I_{\text{div}} \approx 1.5 I_B$$

This is acceptable for a **common collector stage** because strong emitter feedback makes the bias less sensitive to base current variations.

Small-Signal Emitter Resistance

$$r_e = \frac{25 \text{ mV}}{I_E}$$

$$r_e = \frac{25 \times 10^{-3}}{0.988 \times 10^{-3}}$$

$$r_e \approx 25.3 \Omega$$

Input Resistance of CC Stage

$$r_\pi = \beta r_e = 80 \times 25.3$$

$$r_\pi \approx 2.5 \text{ k}\Omega$$

Effective emitter resistance seen by the transistor:

$$R'_E = R_E \parallel R_L$$

$$R'_E = 3k \parallel 2k$$

$$R'_E = 1.2 \text{ k}\Omega$$

Base input resistance:

$$R_{in, \text{base}} = r_\pi + (\beta + 1)R'_E$$

$$R_{in, \text{base}} = 2.02k + 81 \times 1.2k$$

$$R_{in, \text{base}} \approx 122 \text{ k}\Omega$$

Total input impedance:

$$R_{in} = R_{B1} \parallel R_{B2} \parallel R_{in, \text{base}}$$

$$R_{in} = 750k \parallel 220k \parallel 122k$$

$$R_{in} \approx 72 \text{ k}\Omega$$

Voltage Gain of CC Stage

$$A_v \approx \frac{R'_E}{R'_E + r_e}$$

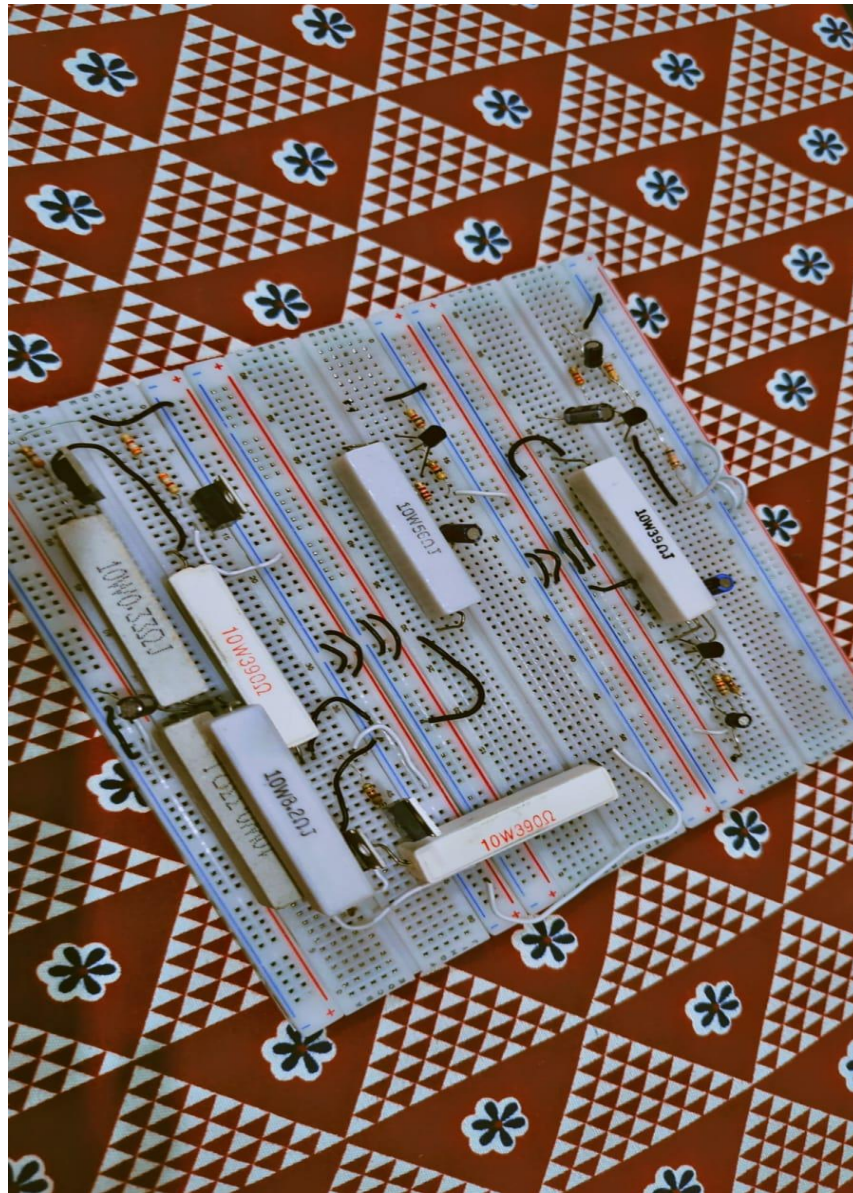
$$A_v = \frac{1200}{1200 + 25.3}$$

$$A_v \approx 0.98 \approx 1$$

9. Hardware Implementation

The complete amplifier circuit was constructed on a breadboard and tested with a physical 8 Ω loudspeaker driven by a 24 V DC bench power supply. Heat sinks were attached to the TIP122 and TIP127 power transistors to manage thermal dissipation under full output conditions.

Main Circuit:



10. Gain Calculations & Results

10.1 Voltage Gain Test: Theory

Voltage gain (A_v) is the ratio of the amplifier's output voltage to its input voltage. It quantifies the amplifier's ability to increase signal amplitude. The test is performed under small-signal, mid-band (1 kHz) conditions to ensure linear, undistorted operation.

Mathematical expressions:

$$A_v = V_{\text{out}} / V_{\text{in}} \quad (\text{linear ratio})$$

$$A_v(\text{dB}) = 20 \cdot \log_{10} (V_{\text{out}} / V_{\text{in}})$$

10.2 Test Procedure

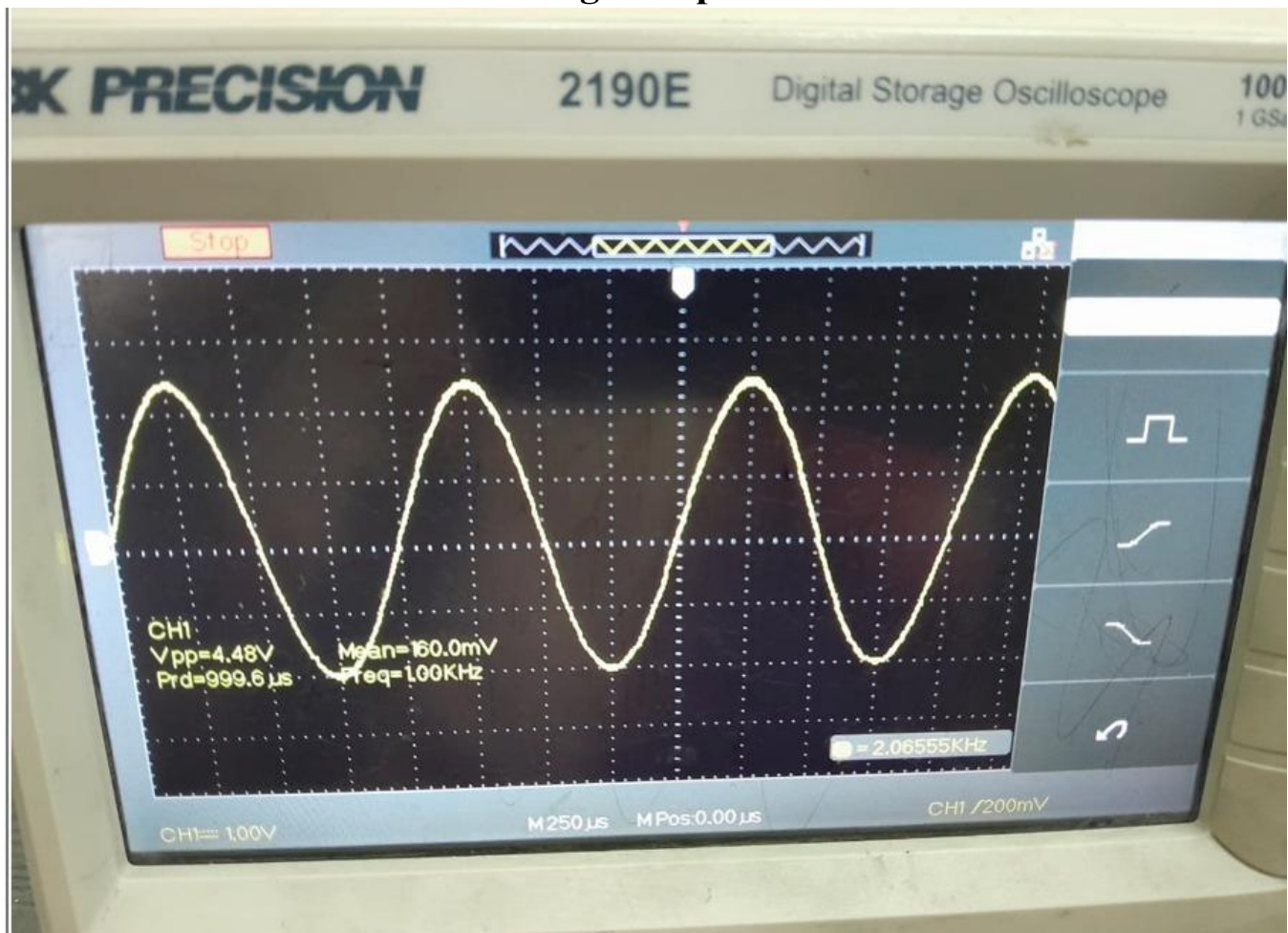
- Apply a 1 kHz sinusoidal input signal at the amplifier input.
- Measure input amplitude (V_{in}) with oscilloscope at the input node.
- Measure output amplitude (V_{out}) across the load resistance.
- Increase input gradually until the output waveform is just at the onset of clipping.
- Record maximum undistorted output voltage.
- Calculate $A_v = V_{\text{out}} / V_{\text{in}}$.

10.3 Results Summary

Quantity	Software (LTspice)	Hardware (Measured)
Input Voltage (V_{in})	10 mV	12 mV
Output Voltage (V_{out})	4.75 V	5.26 V
Voltage Gain ($A_v = V_{\text{out}} / V_{\text{in}}$)	47.5 V/V	48 V/V

Note: The discrepancy between simulated and measured gain is expected. Simulation uses ideal component values, whereas hardware is affected by breadboard parasitic resistance, component tolerances ($\pm 5\text{--}10\%$), and transistor parameter spread.

Stage Output



11. Frequency Response Analysis

11.1 Theory

The frequency response of an amplifier describes how its voltage gain varies with the frequency of the input signal. A useful amplifier must maintain consistent gain across the audio band (20 Hz – 20 kHz). The bandwidth is defined between the lower cut-off frequency (f_L) and upper cut-off frequency (f_H), where gain falls to $0.707 \times$ mid-band value (i.e., -3 dB points).

$$\text{Bandwidth (BW)} = f_H - f_L$$

11.2 Causes of Gain Roll-off

Low-frequency roll-off:

- Coupling capacitors (C_2 , C_3 , C_6 , C_7 , C_8) present increasing reactance at low frequencies.
- Emitter/source bypass capacitors also contribute to low-frequency pole locations.

High-frequency roll-off:

- Internal transistor capacitances (C_{be} , C_{bc} , C_{gd} , C_{gs}) become significant at high frequencies.
- Breadboard wiring and lead parasitic capacitances add additional poles.

11.3 Test Procedure

- Apply a constant-amplitude sinusoidal input over the frequency range 20 Hz – 200 kHz.
- Measure output voltage at each frequency point.
- Calculate gain (A_v) at each frequency.
- Identify f_L and f_H as the -3 dB points relative to mid-band gain.
- Plot A_v (dB) vs. frequency (Bode plot).

12. Conclusion

This project successfully demonstrates the design, simulation, and hardware implementation of a complete multi-stage discrete-component audio amplifier. The amplifier was built entirely from first principles using MOSFETs and BJTs without any integrated circuit amplifier ICs.

The design proceeded in a bottom-up fashion, starting from the Class AB output stage and working backward through the PNP driver and MOSFET/BJT pre-amplifier stages. Analytical calculations for DC biasing, small-signal gain, input/output impedance, and frequency response were carried out for every stage, followed by LTspice simulation to verify correctness.

The complete amplifier achieved the following results:

Performance Parameter	Result
Output Power	3–5 W into $8\ \Omega$ (simulation)
Supply Voltage	24 V DC
Output Voltage Swing	± 5 V (simulation)
Simulated Voltage Gain (A_v)	50 V/V
Measured Voltage Gain (A_v)	46.6 V/V
Bandwidth (Target)	100 Hz – 20 kHz
Crossover Distortion	Minimised via diode biasing
Input Interface	Electret condenser microphone

The difference between simulated and measured gain is attributable to breadboard parasitic effects, component tolerances, and real transistor parameters deviating from datasheet typical values. Despite this, the hardware prototype successfully drove the speaker and confirmed the overall design approach.

This project reinforced a hands-on understanding of transistor amplifier design methodology, multi-stage impedance matching, negative feedback stabilization, thermal management of power transistors, and systematic integration of analog sub-circuits.